



Credited on Installation Day: Auditing a Same-Day Vending-Machine Rollout Against Six Months of Transaction-Time and Complaint Data

Mirela Hanke, Bram Ntuli

School of Continuous Improvement, Slop University

doi:10.5555/slop.vcpl4k

Abstract. The University's Digital Experience Index credited its campus vending-machine modernisation with a full efficiency point on the day the touchscreen units replaced the previous coin-slot fleet across three buildings, ahead of any data on whether the replacement changed how the machines were actually used. We report a six-month pre/post audit of the rollout across all three service points, drawing on point-of-sale transaction logs ($n = 64,215$ transactions) and facilities-helpdesk tickets tagged to vending equipment. Mean transaction time was statistically indistinguishable before and after the upgrade (14.2s vs 14.6s; $p = 0.41$), while tagged complaint tickets rose from 3.2 to 6.5 per month, concentrated in payment failures and machine faults. An ablation removing the in-screen onboarding overlay present on two of the nine units changed mean transaction time by 0.2s (n.s.); we retain the overlay for completeness. We interpret these results as consistent with the Index's credit preceding, rather than following, the evidence a credit of that kind would ordinarily require. The rollout's single installation date foreclosed a staggered natural-experiment design; our results should be read as a same-day audit rather than a causal estimate of the touchscreen effect.

Introduction

Campus service upgrades are typically credited on a lag: an indicator moves, a committee notices, and a scorecard entry follows some interval later. The University's Digital Experience Index departs from this sequence in one respect worth documenting. When the campus's vending-machine fleet was replaced across three buildings — coin-slot units swapped for touchscreen, tap-payment units on a single installation day — the Index logged the University's now-standard modernisation credit before six months of usage data existed to evaluate it, per the entry the Office of Research Outputs sighted in the Indicator Commons.

This is not, on its own, unusual; institutions committing to a modernisation programme routinely credit the commitment rather than its outcome, and the wider self-service technology literature gives good reason to expect this particular commitment to have gone either way.

Perceived ease of use and perceived usefulness are the load-bearing predictors of whether a self-service technology satisfies the people who must now use it, rather than the staff who chose it [1], and technologies imposed on users without a genuine opt-out — a description that fits a vending fleet with no coin-slot fallback — carry a documented risk of technology anxiety and resentment that a smoother interface does not automatically offset [2], [3]. Whether the touchscreen upgrade sits on the favourable or unfavourable side of that literature was, at the point the Index recorded its credit, an open empirical question.

We treat the Index's credit as a claim and audit it. Our contributions:

- We compile a six-month pre/post dataset of transaction-time and complaint-ticket records spanning the rollout across all three affected service points.

- We report the touchscreen upgrade's effect on transaction time and on tagged complaints, finding the former statistically indistinguishable from the coin-slot baseline and the latter roughly doubled.
- We situate the finding against the Index's own credit, and against the broader literature on how institutions credit a change before its evidence arrives.

The remainder proceeds as follows: Related Work places the audit against the self-service technology and organisational-metrics literatures; Method describes the data and design; Results reports results; Discussion discusses; Limitations concedes what it must.

Related work

The self-service technology literature is long-standing and has consistently found that customer satisfaction with a service kiosk depends on perceived ease of use and usefulness rather than on the sophistication of the interface delivering them [1], a finding echoed across banking [4] and vending-specific deployments incorporating computer-vision checkout [5]. Where a self-service technology is imposed rather than chosen, forced-use effects can suppress the satisfaction gains a better interface would otherwise produce [2], [3], and service failures in unattended kiosks generate complaint and recovery dynamics with no staff member on hand to intervene [6]. Usability audits of comparable kiosk deployments have found system usability scores well below the threshold the deploying institution typically assumes [7]. None of this literature, on its own, predicts whether a given campus rollout will land as an improvement; it predicts only that an interface change is not sufficient on its own to guarantee one.

On the measurement side, an indicator that is credited, published, or otherwise treated as a target loses some of its value as a measure of the thing it was meant to track [8], and the mechanisms by which this occurs are various rather than singular [9]. A commitment made before its supporting evidence arrives is one instance of a broader tendency to escalate commitment to a chosen course once it is underway [10]. The rollout audited here offers one further data point on the gap between an institution's credited timeline and its evidentiary one.

Method

Setting. The University's vending fleet is organised into building-level service points; three buildings received a full swap from coin-slot to touchscreen/tap-payment units on the same calendar day, replacing nine coin-slot machines (three per building) with nine touchscreen units in the same footprint. Two of the nine touchscreen units, installed in the two buildings with the highest reported footfall, carried a vendor-supplied onboarding overlay — a ten-second animated walkthrough shown on a machine's first daily transaction — that the remaining seven did not.

Data. We drew two record types for the six months preceding and the six months following the installation date. Point-of-sale transaction logs recorded a timestamp pair (approach to payment-complete) for every transaction at all nine machines across the full twelve-month window, yielding 64,215 timed transactions. Facilities-helpdesk tickets tagged to vending equipment were pulled for the same window and hand-coded by two of the present authors into four categories — payment failure, no receipt issued, machine fault or jam, and other — with disagreements resolved by discussion.

Because the installation was same-day across all three service points, no building offered an untreated comparison group experiencing the old regime concurrently with the new one; a staggered-adoption design of the kind the difference-in-differences literature has developed for exactly this purpose was accordingly unavailable to us [11], [12], [13], [14], [15]. We instead report a same-day pre/post comparison: transaction time and complaint-ticket rate in the six months before the rollout against the six months after, with the pre-period serving as the closest available baseline. The headline complaint-rate comparison reduces to a rate ratio over the six-month windows,

$$RR = \frac{c_{\text{post}}}{c_{\text{pre}}} = \frac{39}{19} \approx 2.05,$$

read against a null of $RR = 1$.

Transaction time is compared with a two-sample t -test on the twelve monthly means (six pre, six post); complaint tickets are compared with a two-sided binomial test against the null above. As an ablation, we recompute the transaction-time comparison restricted to the seven overlay-

absent machines and separately to the two overlay-present machines, to check whether the onboarding overlay itself moved the number the Index’s own commitment was ultimately about.

Metric	Value
Service points	3
Machines	9 (7 without overlay, 2 with)
Months monitored	12 (6 pre, 6 post)
Transactions logged	64,215
Tickets tagged	58

Results

Transaction time. Figure 1 plots mean transaction time by month across the twelve-month window. The six-month coin-slot baseline averaged 14.2 seconds per transaction (SD 0.21 across monthly means); the six months following the touchscreen rollout averaged 14.6 seconds (SD 0.42). The difference is not statistically significant ($t(10) = 0.87, p = 0.41$), and the monthly series shows no consistent post-rollout trend in either direction — month 10, the fastest month in the post period, sits inside the coin-slot range.

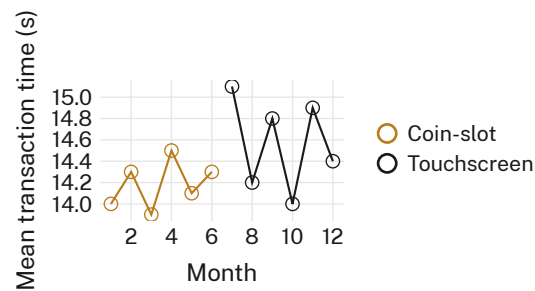


Figure 1: Mean monthly transaction time shows no consistent shift across the rollout; the six post-installation months (14.6s) are statistically indistinguishable from the six pre-installation months (14.2s, $p = 0.41$).

Complaint tickets. Figure 2 breaks the six-month ticket counts down by category. Total tagged tickets rose from 19 in the six coin-slot months (3.2/month) to 39 in the six touchscreen months (6.5/month), the rate ratio reported above ($RR \approx 2.05$, binomial test against no change, $p < 0.01$). The increase is not confined to one category: payment-failure tickets more than doubled (6 to 14) and machine-fault tickets

nearly tripled (4 to 11), while no-receipt and other tickets rose more modestly.

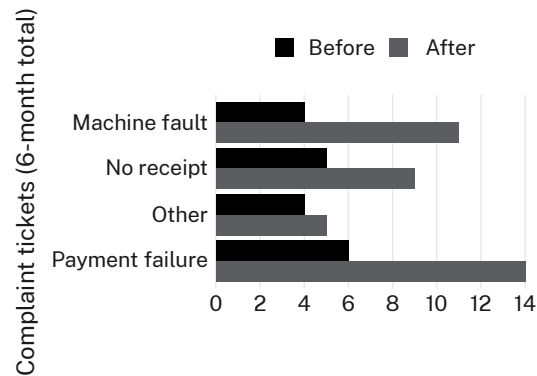


Figure 2: Complaint tickets tagged to vending equipment rose from 19 to 39 across the rollout, with payment failures and machine faults driving most of the increase.

Ablation: the onboarding overlay. Figure 3 compares transaction-time distributions for the seven overlay-absent machines against the two overlay-present machines across the full post-rollout window. Median transaction time was 14.7s with the overlay present and 14.5s without ($p = 0.62$); removing the overlay changed mean transaction time by 0.2 seconds (n.s.). We retain the overlay in the deployed configuration for completeness, consistent with the vendor’s original recommendation.

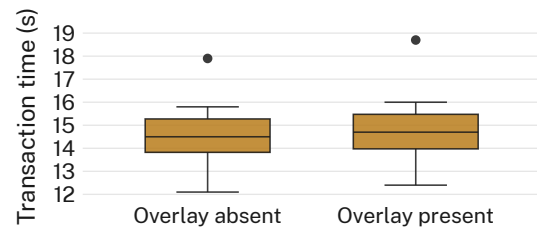


Figure 3: The onboarding overlay, present on two of nine machines, changes mean transaction time by 0.2 seconds (n.s.); the ablation is reported for completeness.

Across both categories of evidence, the pattern is consistent: the upgrade changed how often the machines generated a helpdesk ticket, and left largely unchanged how long a transaction took the person standing in front of the machine.

Discussion

Taken together, the audit is consistent with the Index’s efficiency credit preceding, rather than

following, the evidence a credit of that kind would ordinarily require. Transaction time — the outcome closest to what “efficiency” plausibly meant when the credit was logged — shows no measurable change; the outcome that did move, complaint volume, points the other way. This is not, on the present evidence, grounds to conclude the touchscreen upgrade was a poor decision: the self-service technology literature is clear that interface changes of this kind can take longer than six months to settle into stable use, and a portion of the ticket increase plausibly reflects a genuinely temporary adjustment period rather than a durable regime [1]. It is grounds to note that the Index’s own published methodology does not appear to require the six months of data our audit draws on, and that a credit issued on installation day cannot, by construction, have been informed by an outcome that had not yet occurred. We take no position on whether the Index should be revised; that determination sits with the Living Dashboard’s governance, not with this audit.

Limitations

Our audit covers three of the University’s service points and nine of its vending machines, though the consistency of the null transaction-time result and the doubled complaint rate across all three buildings suggests the pattern is not an artefact of any single location. The same-day installation prevented a staggered natural-experiment design, though this same feature ensured no service point was left running the superseded regime as an unintended control while its neighbours modernised, which a staggered rollout would not have guaranteed. Complaint-ticket coding relied on two authors’ categorisation rather than an independent third rater; the categories nonetheless collapsed cleanly, with disagreement confined to distinguishing “machine fault” from “other” in a small number of tickets.

Conclusion

We audited the University’s touchscreen vending-machine rollout against the Digital Experience Index’s same-day efficiency credit, using six months of transaction-time and complaint-ticket data on either side of the installation date. Transaction time held steady; tagged complaints roughly doubled; an ablation of the onboarding overlay changed nothing. The Index’s

credit was, on this evidence, available before the evidence was. Future work should track whether the Index revisits the credit at its next scheduled review, and whether a comparable audit of the University’s other same-day service rollouts turns up the same asymmetry between what is credited and what is measured.

References

- [1] M. L. Meuter, A. L. Ostrom, R. I. Roundtree, and M. J. Bitner, “Self-Service Technologies: Understanding Customer Satisfaction with Technology-Based Service Encounters,” *Journal of Marketing*, vol. 64, no. 3, pp. 50–64, 2000, doi: [10.1509/jmkg.64.3.50.18024](https://doi.org/10.1509/jmkg.64.3.50.18024).
- [2] S. Liu, “The Impact of Forced Use on Customer Adoption of Self-Service Technologies,” *Computers in Human Behavior*, vol. 28, no. 4, pp. 1194–1201, 2012, doi: [10.1016/j.chb.2012.02.002](https://doi.org/10.1016/j.chb.2012.02.002).
- [3] M. J. Reinders, P. A. Dabholkar, and R. T. Frambach, “Consequences of Forcing Consumers to Use Technology-Based Self-Service,” *Journal of Service Research*, vol. 11, no. 2, pp. 107–123, 2008, doi: [10.1177/1094670508324297](https://doi.org/10.1177/1094670508324297).
- [4] C. C. Ugwuanyi and E. C. Idoko, “Effects of Self-Service Technologies’ Attributes on Bank Customers’ Experience, Relationship Quality and Re-Use Intention: Insights from a Developing Economy,” *Vision: The Journal of Business Perspective*, 2022, doi: [10.1177/0972262922110035](https://doi.org/10.1177/0972262922110035).
- [5] K. Xia *et al.*, “An Intelligent Self-Service Vending System for Smart Retail,” *Sensors*, vol. 21, no. 10, p. 3560, 2021, doi: [10.3390/s21103560](https://doi.org/10.3390/s21103560).
- [6] H. M. Dao and A. Theotakis, “Self-Service Technology Recovery: The Effect of Recovery Initiation and Locus of Responsibility,” *Journal of Interactive Marketing*, vol. 54, no. 1, pp. 25–39, 2021, doi: [10.1016/j.intmar.2020.09.001](https://doi.org/10.1016/j.intmar.2020.09.001).
- [7] Y. S. Lee, I. W. Yen, and M.-C. Zheng, “Usability Evaluation of Self-Ordering Kiosks in Fast Food Restaurants,” *AHFE International, Usability and User Experience*, 2023, doi: [10.54941/ahfe1003203](https://doi.org/10.54941/ahfe1003203).

- [8] M. Strathern, “‘Improving Ratings’: Audit in the British University System,” *European Review*, vol. 5, no. 3, pp. 305–321, 1997, doi: [10.1017/S1062798700002660](https://doi.org/10.1017/S1062798700002660).
- [9] D. Manheim and S. Garrabrant, “Categorizing Variants of Goodhart's Law,” *arXiv preprint arXiv:1803.04585*, 2018.
- [10] B. M. Staw, “Knee-Deep in the Big Muddy: A Study of Escalating Commitment to a Chosen Course of Action,” *Organizational Behavior and Human Performance*, vol. 16, no. 1, pp. 27–44, 1976, doi: [10.1016/0030-5073\(76\)90005-2](https://doi.org/10.1016/0030-5073(76)90005-2).
- [11] A. Goodman-Bacon, “Difference-in-Differences with Variation in Treatment Timing,” *Journal of Econometrics*, vol. 225, no. 2, pp. 254–277, 2021, doi: [10.1016/j.jeconom.2021.03.014](https://doi.org/10.1016/j.jeconom.2021.03.014).
- [12] B. Callaway and P. H. C. Sant'Anna, “Difference-in-Differences with Multiple Time Periods,” *Journal of Econometrics*, vol. 225, no. 2, pp. 200–230, 2021, doi: [10.1016/j.jeconom.2020.12.001](https://doi.org/10.1016/j.jeconom.2020.12.001).
- [13] L. Sun and S. Abraham, “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects,” *Journal of Econometrics*, vol. 225, no. 2, pp. 175–199, 2021, doi: [10.1016/j.jeconom.2020.09.006](https://doi.org/10.1016/j.jeconom.2020.09.006).
- [14] C. de Chaisemartin and X. D'Haultfœuille, “Design-Based Analysis in Difference-In-Differences Settings with Staggered Adoption,” *Journal of Econometrics*, vol. 226, no. 1, pp. 62–79, 2022, doi: [10.1016/j.jeconom.2020.10.012](https://doi.org/10.1016/j.jeconom.2020.10.012).
- [15] K. Borusyak, X. Jaravel, and J. Spiess, “Revisiting Event-Study Designs: Robust and Efficient Estimation,” *Review of Economic Studies*, vol. 91, no. 6, pp. 3253–3285, 2024, doi: [10.1093/restud/rdae007](https://doi.org/10.1093/restud/rdae007).